Program: **Diploma in Food Processing Technology**	
Course Code: **5423C**	Course Title: **Beverage Processing**
Semester: **5**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4(L: 4 T:0 P: 0)**	Periods per semester:**60**

Course Objectives:

1. To understand the basic ingredients used in beverages
2. To study the production process of beer, wine and various other beverages
3. To impart knowledge on the quality parameters of beverages

Course Prerequisites:

Topic	Program/CourseName
Basic knowledge of beverages	Beverage processing

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe the basic ingredients and classification of beverages.	13	Understanding
CO2	Explain the key ingredients and the beer manufacturing process, including malting, brewing, fermentation, pasteurization, and packaging, as well as describe the fermentation types in wine.	17	Understanding
CO3	Describe the process techniques and equipment used for the manufacture of distilled fermented beverages	17	Understanding
CO4	Explain the types, manufacturing processes, quality controls, and safety regulations related to packaged drinking water and beverages.	13	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Describe the basic ingredients of beverages and classification**		
M1.01	Define Beverage and its classification	3	Understanding
M1.02	Discuss about Sweeteners and flavoring agents	5	Understanding
M1.03	Describe about Colors, flavors Preservatives, Emulsifiers and Stabilizers in beverages	5	Understanding
Contents: Beverage definition, classification-ingredients-water, CO_2, Bulk Sweeteners, flavoring agents, Colors-natural and artificial, preservatives, Emulsifiers and Stabilizers			
CO2	**Explain the key ingredients and the beer manufacturing process, including malting, brewing, fermentation, pasteurization, and packaging, as well as describe the fermentation types in wine.**		
M2.01	Discuss the various Ingredients in beer	4	Understanding
M2.02	Explain Beer manufacturing process	7	Understanding
M2.03	Discuss on types of Wine fermentation	5	Understanding
	Series Test –I	1	

Contents: Ingredients-Malt, Hops-adjuncts-water, yeast. Beer manufacturing process-malting, preparation of sweet wort, brewing, fermentation, pasteurization and packaging. Wine fermentation-types-red and white			
CO3	**Describe the process techniques and equipment used for the manufacture of distilled fermented beverages**		
M3.01	Discuss Malt and Grain Whisky-types, raw materials and processing	7	Understanding
M3.02	Outline the Design of continuous distillation	7	Understanding
M3.03	Summarize the Whiskies of the world and their regulations	3	Understanding
Contents: Malt and Grain, Whisky-types, raw materials and processing. Design-continuous distillation stills, Whiskies of the world and their regulations.			
CO4	**Explain the types, manufacturing processes, quality controls, and safety regulations related to packaged drinking water and beverages.**		
M4.01	Define Packaged drinking water	3	Understanding
M4.02	Explain the effective application of quality controls	5	Understanding
M4.03	Review Food safety regulations	4	Understanding
	Series Test–II	1	
Contents: Packaged drinking water- definition, types, manufacturing processes, Effective application of quality controls-Sanitation and hygiene in beverage Industry-quality of water used in beverages- threshold limits of various ingredients –Food safety regulations-Requirements of Soluble Solids and Titrable Acidity in Beverages			

Text / Reference

T/R	Book Title/Author
R1	Ashurst,P.R ,"Chemistry and Technology of Soft Drinks and Fruit Juices",2nd Edition, Blackwell Publishing, 2005
R2	Steen,D.P and Ashurst,P.R,"Carbonated soft drinks-Formulation and Manufacture", Blackwell Publishing, 2000..

R3	ShahkunthalaManay,V and Shadakdharaswamy,M, “Food-Facts and Principles”, New Age International Pvt. Ltd, 3rd Revised Edition,2000
R4	Amalendu Chakraverty et al, ”Handbook of post harvest technology”, Ed: Marcel Dekker Inc, (Special Indian Edition), 2000.
R5	Robert W Hukins, ”Microbiology and Technology of fermented Foods”, IFT Press, Blackwell Publishing Ltd. 2006
R6	Charles,W, Bamforth, ”Food, Fermentation and Microorganisms,” Blackwell Science Publishing Ltd., 2005
R7	Inge Russel, Graham Stewart and Charlie Bamforth, ”Whisky-Technology, production and marketing”, Elsevier.2003

Online Resources

Sl.No	**Website Link**
1	https://www.britannica.com/topic/beer
2	https://winefolly.com/deep-dive/common-types-of-wine/
3	https://winefolly.com/deep-dive/common-types-of-wine/
4	https://sensorex.com/three-main-types-of-water-quality-parameters-explained/